

SentiSing: A Real-Time Affective Feedback System for Immersive Karaoke in Virtual Reality

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Figure 1: (Left) A user interacting with the system; (Middle and Right) Real-time affective visual feedback in the VR forest.

ABSTRACT

This paper presents a VR karaoke system that treats voice as a continuous affective input for immersive environmental feedback. Vocal and musical affect are analyzed as joint control signals, mapped to generative visual dynamics in a virtual forest. Arousal and valence function as expressive parameters governing growth, color, and atmospheric change. Users perceive how variations in vocal expression and musical emotion shape evolving environmental responses in real time. This work explores the design space of affect-driven VR music interaction, showing how coupled vocal and musical affect can support expressive engagement in immersive singing. Initial observations suggest the system prompts users to actively experiment with their vocal expression in response to the song's emotional content.

Index Terms: Virtual Reality, Vocal Expression, Affective Computing, Immersive Music Systems.

1 INTRODUCTION

Singing is a highly embodied activity in which vocal expression unfolds dynamically over time. Beyond pitch and loudness, the human voice carries rich affective variation that reflects expressive intensity and emotional nuance [3]. However, in many digital karaoke and music interaction systems, singing is primarily supported through auditory output or symbolic visual cues, such as pitch indicators and scoring, while the affective dimension is rarely considered and immersive emotional feedback remains limited.

Virtual reality offers a space to reconsider what singing interfaces might respond to. Affective vocal expression remains an underexplored interaction input in immersive systems. Singing provides a continuous stream of emotional modulation that users intuitively shape with music. Treating vocal affect as interaction ma-

terial suggests a way for VR environments to respond not only to action, but to felt experience through multisensory feedback.

This work presents SentiSing, a VR singing system that couples vocal and musical affect with immersive environmental feedback. By spatializing emotional dynamics as visual responses, the system enables singers to experience their affective expression as perceptible environmental change and explore its relationship to the song's emotional flow.

2 RELATED WORK

Research in the NIME and interactive music communities has established sound features as effective continuous control parameters [4], demonstrating that expressive and learnable interaction can emerge when sonic qualities directly modulate system behavior. Studies on virtual singing further show that voice-based interaction in VR enhances presence and embodied experience by situating vocal activity within spatial and multisensory environments [5]. Such immersive feedback allows performers to engage more deeply with both their voice and the accompanying music.

In parallel, affective interaction research has increasingly adopted continuous arousal-valence models to represent emotional dynamics over time [3, 6]. Within music interaction systems, these models enable visual feedback to respond fluidly to expressive variation. Color, validated as a salient affective cue, is commonly used to convey emotional valence through warm-cool mappings [1]. In this process, real-time visualization supports performers' ongoing perception and modulation of their own vocal expression, forming a continuous affective feedback loop [2].

Building on these lines of work, how vocal input, affective analysis, and multisensory feedback can be jointly explored as an integrated affective experience in immersive VR singing remains largely unexplored. This project investigates this space by examining affect-driven singing interaction in VR, including the combined influence of vocal and musical affect on emotional engagement.

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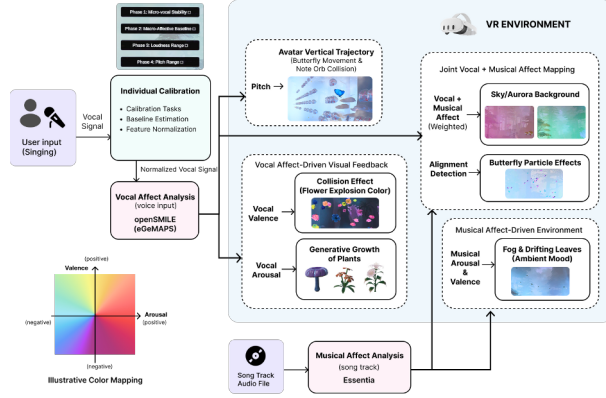


Figure 2: System pipeline

3 METHODOLOGY

3.1 System Implementation and Affect Analysis

The prototype is implemented in Unity 6 and deployed on an Oculus Quest 3, with vocal input captured via a dynamic microphone and analyzed in real time.

To drive the virtual environment, the system integrates two complementary data streams represented within a shared continuous arousal–valence space. Vocal affect is analyzed in real time using openSMILE (eGeMAPS feature set), acting as a direct control signal that reflects moment-to-moment expressive variation. Conversely, musical affect is pre-analyzed using Essentia to capture the song’s evolving emotional contour, which modulates the overall atmospheric context.

3.2 Individual Calibration

Before singing begins, the system performs an individual calibration phase to establish a personalized acoustic reference. Users complete controlled vocal tasks (such as sustained phonation and variations in pitch and intensity) to capture the statistical distribution of their vocal features in a singing state. This reference normalizes subsequent arousal and valence outputs, enabling affective features to be interpreted as relative deviations from the user’s baseline rather than absolute emotional states. Combined with confidence gating based on signal quality, the calibration stabilizes real-time interaction while preserving individual differences in vocal range and expressive style.

3.3 Interaction

Users are situated in a fantasy forest environment where singing gradually transforms the surrounding scene through metaphorical representations of affect in color, growth activity, and atmospheric motion, making both vocal expression and musical emotion perceptible in space.

Interaction is driven by singing. Pitch controls the butterfly’s vertical movement and its interaction with MIDI-generated note orbs, providing real-time feedback on vocal activity. Through the interplay of vocal and musical affect, visual feedback evolves during singing, situating individual vocal expression within the song’s emotional landscape.

3.4 Affect-Driven Environmental Mapping

Affective features modulate multiple visual elements within the forest scene. Vocal valence influences the color of flower bloom effects that appear when the butterfly passes through note orbs, using warm and cool tones to convey affective polarity. Vocal arousal

controls the density of magical vegetation emerging along the butterfly’s flight path, linking expressive intensity to visible growth activity within the environment.

Musical affect modulates surrounding fog and leaf coloration to establish an ambient emotional atmosphere that mirrors the song’s mood dynamically. The broader background sky and aurora colors are jointly controlled by vocal and musical affect, with greater weight assigned to vocal expression to emphasize real-time performance nuances. To highlight voice-music affective alignment, the system triggers subtle particle effects as non-scoring visual cues when vocal and musical affect coincide within predefined regions of the arousal–valence space. These cues invite singers to sense moments when their expression aligns with the song’s emotional flow, supporting a felt experience of resonance.

3.5 Pilot Study

After completing the prototype, six participants took part in an initial pilot experience involving individual calibration and the full singing interaction. Feedback indicated that the environmental response made vocal expression perceptible, prompting users to actively experiment with and adapt their emotional tones. Participants described using the visual feedback to consciously harmonize their delivery with the evolving scenery and the song’s atmosphere. These observations provide initial support that affect-driven visual feedback can mediate vocal expression through immersive interaction, fostering emotional engagement.

4 CONCLUSION AND FUTURE WORK

This work explores the integration of vocal and musical affect as dual control signals for shaping multimodal singing experiences in VR. Preliminary observations suggest that visualizing these affective relationships encourages users to engage more deeply with their emotional expression and align it with the music’s mood.

Future work will focus on improving system robustness and reducing real-time latency, followed by larger-scale user studies to more systematically examine how the interaction influences different aspects of the singing experience, including perceived emotional resonance with the music. Building on these findings, future research will further refine affect-driven visual mappings and voice–music alignment mechanisms informed by cognitive psychology. Additional directions include extending visual mappings to richer vocal and musical features such as timbre, vocal style, and rhythm, and investigating how affect-to-visual mappings may vary across musical genres and cultural contexts.

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